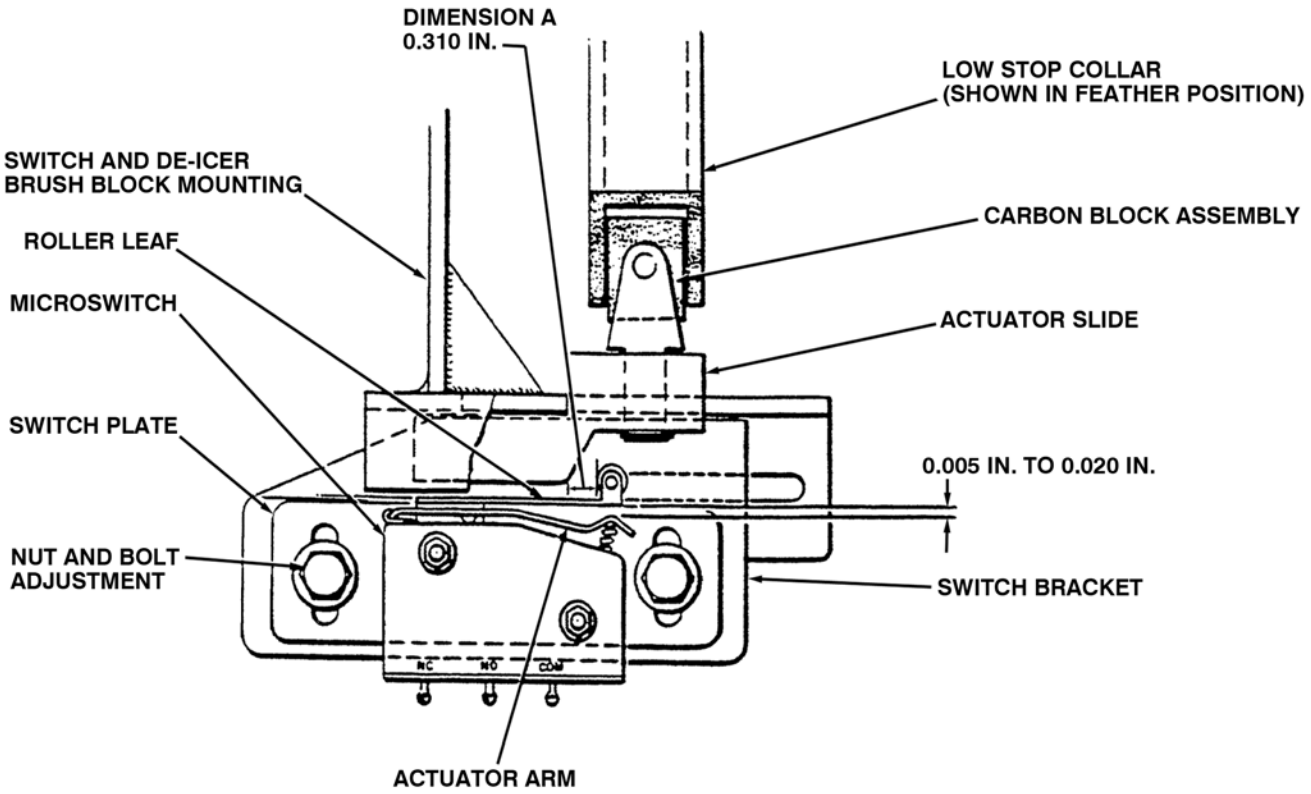
Beta Microswitch Assembly (Mod 6/2022) – General Data

1. General

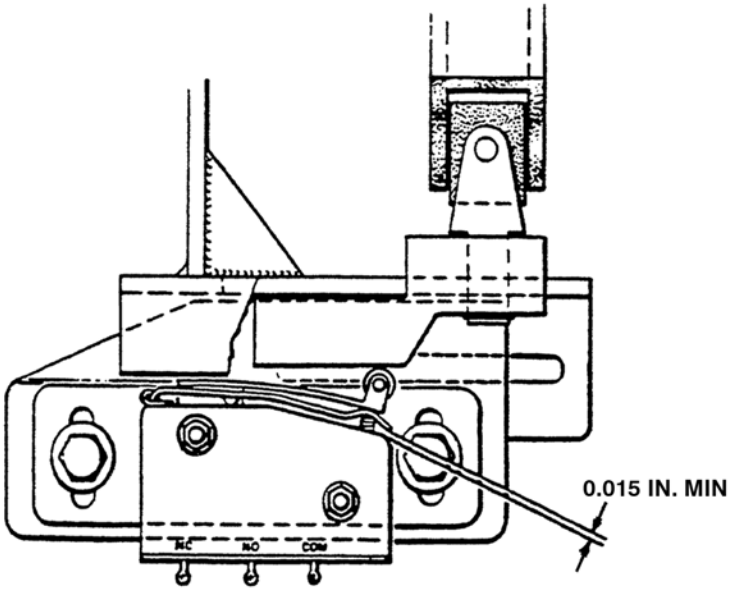
Refer to Figure 1 for details.

The beta microswitch assembly comprises a switch and de-icer brush block mounting, switch bracket, switch plate, microswitch, actuator slide, and carbon block assembly. The switch and de-icer brush block mounting, secured to the engine reduction gear case by two nuts, incorporates a slot in which operates the actuator slide, and two mounting faces. During manufacture, the mounting and actuator slide are treated with a dry film lubrication process (Electrofilm), making it unnecessary to lubricate these parts in service. One of the mounting faces supports the propeller de-icing brush block assembly (when fitted refer to 30-60-11, Propeller De-icing Brush Module Assembly (Mod 6/1674) – Maintenance Practices), and the other the switch bracket. The switch bracket, secured to the mounting with two bolts, washers and anchor nuts, has a lengthwise slot, and the switch plate, bolted to the switch bracket, has two slots across its width. The slot in the switch bracket allows the microswitch operating position to be adjusted, and the slots in the switch plate permit correct microswitch engagement with actuator slide.

During operation the carbon block assembly, actuated by the propeller low stop collar, moves the actuator slide which operates the microswitch when the propeller blades reach an angle of +9 degrees. The microswitch contacts are normally open above, or closed below +9 degrees. A propeller blade movement of 1 degree moves the actuator slide approximately 0.030 inch, during the range in which the low stop collar operates.



WITH PROPELLER BLADES AT ZERO PITCH AND SWITCH OPERATED, CHECK THAT AN OVERTRAVEL, CHECK THAT AN OVERTRAVEL CLEARANCE OF 0.015 IN. MIN EXISTS BETWEEN ROLLER AND SLIDE



Beta Microswitch Assembly
Figure 1

2. Removal/Installation

A. Remove Beta Microswitch

1) Gain access to front of engine reduction gear case.

2) Remove and retain the two bolts and washers securing switch plate to switch bracket.

3) Carefully remove the heatshrink from the switch.

4) Disconnect microswitch wires at the switch by removing the solder in accordance with 20-73-00, Soldering of Electrical Terminals – General Data.

5) Remove the two nuts, screws and washers securing microswitch to switch plate. Retain nuts, screws, washers and switch plate.

B. Install Beta Microswitch

1) Secure microswitch to switch plate, using plate, screws, washers and nuts retained during microswitch removal.

2) Slide the heatshrink onto the two beta microswitch wires.

3) Solder the two wires to the microswitch terminals in accordance with 20-73-00, Soldering of Electrical Terminals – General Data.

Note The wires attaching to the microswitch are nickel plated and require a specific type of solder. Refer to 20-73-00, Soldering of Electrical Terminals – General Data for further information.

4) Locate switch plate, complete with microswitch, on switch bracket and attach using bolts and washers retained during microswitch removal. A washer should be located under the bolt heads and under each nut. Do not fully tighten bolts at this stage.

5) Adjust microswitch; refer to Para 3.A.

6) Secure access panels as necessary.

C. Remove Beta Microswitch and Propeller De-icing Brush Block Assembly

1) Remove propeller de-icing brush block (refer to 30-60-11, Propeller De-icing Brush Module Assembly (Mod 6/1674) – Maintenance Practices), if fitted.

2) Carefully remove the heatshrink from the switch.

3) Disconnect microswitch wires at the switch by removing the solder in accordance with 20-73-00, Soldering of Electrical Terminals – General Data.

4) This may be achieved as in Para 2.C.4).a) or Para 2.C.4).b) as follows:

a) Using retraction clamp (SD12559) as in propeller installation (refer to 61-00-00, Propellers – General Data).

Note When using the retraction clamp, strictly observe the procedures regarding installation and use.

b) Turn the propeller blades into reverse pitch and secure in this position.

Note To facilitate turning of blades, remove propeller piston flexlock nut, using wrench (SD12549).

5) Remove the two nuts securing switch and de-icer brush block mounting to reduction gearbox. Remove mounting, complete with actuator slide, carbon block, switch bracket, switch plate and microswitch. Retain nuts.

D. Install Beta Microswitch and Propeller De-icing Brush Block Assembly

1) Before installation, check that the actuator slide moves freely in slot in switch bracket. If a new carbon block is being fitted, the following points should be noted:

a) No tight spots exist when sliding block around groove in low stop collar. If necessary, lightly sand block to give a clearance of 0.001 to 0.002 inch at tightest point.

b) Lacquer has been removed from stem of block.

c) Cotter pins retaining carbon block in axle are tightly fitted and trimmed to prevent fouling on low stop collar.

d) Ensure carbon block swivels freely in actuator slide; lightly lubricate carbon block stem and each end of axle with one drop of engine oil.

2) Turn propeller blades into reverse pitch.

This may be achieved as in Para 2.D.2).a) or Para 2.D.2).b) as follows:

a) Using retraction clamp (SD12559) as in propeller installation (refer to 61-10-00, Propeller Assembly – General Data).

Note When using the retraction clamp, strictly observe the procedures regarding installation and use.

b) Turn the propeller blades into reverse pitch and secure in this position.

Note To facilitate turning of blades, remove propeller piston flexlock nut, using wrench (SD12549).

3) Using the two nuts retained during mounting removal, install mounting complete with actuator slide, carbon block, switch plate and microswitch. Ensure carbon block engages low stop collar. Torque nuts securing mounting to engine reduction gearbox to between 145 to 165 pound–inches.

4) Slide the heatshrink onto the two beta microswitch wires.

5) Solder the two wires to the microswitch terminals in accordance with 20-73-00, Soldering of Electrical Terminals – General Data.

Note The wires attaching to the microswitch are nickel plated and require a specific type of solder. Refer to 20-73-00, Soldering of Electrical Terminals – General Data for further information.

6) Adjust microswitch; refer to Para 3.A.

7) Install propeller de-icing brush block assembly (refer to 30-60-11, Propeller De-icing Brush Module Assembly (Mod 6/1674) – Maintenance Practices).

3. Adjustment/Test

Refer to Figure 1 for details.

A. Adjust Beta Microswitch

1) Slacken bolts securing switch plate to switch bracket.

2) Check low stop collar runout (refer to 61-10-00, Propeller Assembly – General Data).

3) Check that a clearance of 0.005 to 0.020 inch exists between roller leaf and actuator arm when roller is not contacting slide.

4) Position propeller low stop collar so that microswitch roller is positioned on operated face of slide. This may be achieved as in Para 3.A.4).a) or Para 3.A.4).b) as follows:

   Note When using the retraction clamp, strictly observe the procedures regarding installation and use.

   a) Using retraction clamp (SD12559) as in propeller installation (refer to 61-10-00, Propeller Assembly – General Data.

      Note To facilitate turning of blades, remove propeller piston flexlock nut, using wrench (SD12549).

   b) Turn the propeller blades into zero pitch and secure in this position. If zero thrust stop latches are fitted, turn blades until latches engage.

5) Move switch plate assembly upwards against slide face until switch actuates (audible click), then tighten retaining bolts.

6) Check gap between actuator arm and microswitch body is not less than 0.015 inch.

7) Position low stop collar to feathered position using Para 3.A.7).a) or Para 3.A.7).b) as follows:

   Note When removing retraction clamp, observe correct procedures for replacement of rod end ring.

   a) Remove retraction clamp and replace rod end ring as in propeller installation (refer to 61-10-00, Propeller Assembly – General Data).

   b) Turn propeller blades to the feathered position. If zero thrust stops are fitted, release latches before attempting to turn blades. Install flexlock nut, ensuring O-ring seal is in position, and torque to 120 pounds feet using wrench (SD12549).

8) Slacken bolts securing switch bracket to mounting.

9) Adjust clearance between microswitch roller and actuator slide (dimension A) to 0.310 inch, utilizing the switch bracket slot.

10) Tighten bolts securing switch bracket to mounting.

11) Recheck clearance.

12) Complete ground run and check for beta microswitch operation (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

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